Louis Bensard

Co-founder & CTO · Research & Applied AI Engineer
New York, NY · louis.bensard@gmail.com · (646) 244-7613
louisbensard.com · linkedin.com/in/louis-bensard · github.com/LouisBensard

SUMMARY

AI engineer with 7 years of experience training deep learning models and deploying them in production for enterprise customers. As co-founder & CTO of Sightline and its first engineer, built its AI system end to end: a transformer trained from scratch in PyTorch, the evaluation framework that gates every model change, the recommendation engine on top of it and the GPU training and data infrastructure behind it, and led every client deployment through go-live. Previously 5 years shipping ML for Fortune 500 clients and leading a team of 9+ engineers.

EXPERIENCE

SIGHTLINE New York, NY
Co-founder & CTO (previously Founding Head of AI & ML) *Aug 2025 – Present*
First engineer. Built the company's AI system, recommendation engine, data platform, engineering team from scratch, now in production with enterprise customers.

- **Transformer model:** Designed and implemented a PatchTST transformer from scratch in PyTorch (patch tokenization, learned positional and per-series embeddings, multi-head encoder) with a hurdle architecture for zero-inflated targets. Replaced statistical baselines; sub-5% wMAPE across thousands of series.
- **Training and evaluation:** Built the training stack: AdamW with LR scheduling and early stopping, 5-seed training with top-k ensembling, point-in-time backtesting, and an evaluation framework that gates every model change.
- **GPU training platform:** Built model training on AWS Batch: CUDA images in ECR, scheduled retraining via Lambda, versioned model artifacts in S3, inference-only daily runs and experiment tracking in Weights & Biases.
- **AI decision engine (SCAI):** Invented and built Supply Continuity AI, a recommendation engine new to the category that turns model outputs and inventory projections into scored escalations with a next-best action, recommended order quantities and dates.
- **Client implementations:** Technical lead for all our enterprise implementations end to end: scoped data integrations with each client's distributors and IT teams, built the connections and mappings, and managed the lifecycle through go-live.
- **Data platform:** Designed the event-driven ingestion platform (AWS Transfer Family, S3, SQS, Lambda, ECS, EC2): a config-driven mapper normalizing data into one canonical schema, idempotent loads into PostgreSQL, and Grafana monitoring with alerting. Millions of rows processed daily.
- **Product and team:** Co-architected the multi-tenant product (Next.js, TypeScript, Prisma, Vercel) with org isolation and passwordless RDS access via IAM; built the engineering team and standards: hiring, code review, testing, CI/CD and documentation.

Tech: Python, PyTorch, scikit-learn, pandas, SQLAlchemy, PostgreSQL, AWS (Batch, S3, SQS, Lambda, ECS, Transfer Family, RDS, Secrets Manager), Docker, Weights & Biases, Grafana, GitHub Actions, TypeScript, Next.js, Prisma

EKIMETRICS New York, NY
Tech Lead, Machine Learning *Jul 2024 – Aug 2025*

- **Modeling and deployment:** Built Bayesian marketing-mix models (PyMC, MCMC) and reusable Python/SQL modeling frameworks for Fortune 500 retail and beauty clients, from scoping through deployment; $20M+ in measured client impact.
- **Client delivery and leadership:** Led a team of 9+ engineers; primary technical contact for 10+ strategic accounts, presenting results to C-level executives.

STRATOS AVIATION New York, NY
Founder & Lead AI/ML Engineer *Aug 2023 – Dec 2024*
A "Strava for pilots" iOS app, live on the App Store. Designed, built and shipped solo, from concept to launch in 8 months.

- **LLM fine-tuning:** Curated the training data and fine-tuned an OpenAI model into a pilot assistant that gives flight-performance recommendations.
- **Recommender system:** Built personalized airport and route recommendations from nearest-neighbor retrieval over flight time series and geospatial features across 100K airports and 10M+ points of interest, with on-device inference via Core ML.
- **ML pipelines and security:** Automated preprocessing, feature selection (Boruta, RFE), evaluation and Core ML export; built end-to-end encrypted messaging between pilots (RSA-2048, AES-256).

IPSOS MMA New York, NY
Tech Lead, Machine Learning *Jul 2019 – Jun 2023*
Promoted three times from Data Scientist (ML Engineering) to Tech Lead.

- **Predictive modeling at scale:** Developed and deployed hundreds of models, from gradient boosting to neural networks (XGBoost, TensorFlow, scikit-learn), on 10B+ data points; automated the team's feature selection, Bayesian hyperparameter optimization and evaluation pipelines.
- **Bayesian modeling:** Built Bayesian and hierarchical models (PyMC3, MCMC inference) for Fortune 500 clients.
- **Interpretability:** Explained model behavior to C-suite executives with SHAP, LIME and interpretable models, on projects worth $4–6M a year.

PANASONIC AVIONICS CORPORATION Fullerton, CA
Machine Learning Engineer (internship) *Jan 2019 – May 2019*

- Built an in-flight entertainment recommendation engine (scikit-learn, TensorFlow) predicting passenger viewing preferences with 85%+ accuracy on a 10M-row dataset; led a 7-person team.

EDUCATION

California State University, Fullerton 2019
M.S. Statistics, summa cum laude (4.0 GPA, top of class, Phi Kappa Phi). Coursework: Machine Learning, Deep Learning, Reinforcement Learning, Bayesian Analysis, Probability Theory.

Sorbonne Université, Paris 2017
B.S. Mathematics and B.S. Computer Science (two degrees), summa cum laude.

SKILLS

Deep learning: Transformers built from scratch (attention, positional and learned embeddings), PyTorch, training loops and optimization (AdamW, LR scheduling, early stopping), multi-seed ensembling, evaluation design and backtesting

AI engineering: LLM fine-tuning and training-data curation, evaluation frameworks, retrieval and recommendation systems, production training and inference pipelines

ML infrastructure: PyTorch, GPU training (CUDA, AWS Batch), experiment tracking (Weights & Biases), model versioning, training and inference pipelines, Docker, CI/CD (GitHub Actions)

Data and cloud: Python, SQL, PostgreSQL, pandas, SQLAlchemy, Alembic, AWS (S3, SQS, Lambda, ECS, Batch, Transfer Family, RDS, IAM, Secrets Manager), Grafana

Product: TypeScript, Next.js, React, Prisma, Vercel, Playwright, Swift, Core ML

Customer delivery: Technical discovery and scoping, enterprise data integrations, implementation through go-live, C-level presentations

Mathematics: Probability theory, linear algebra, Bayesian statistics, optimization, statistical modeling

PASSIONS

Licensed private pilot · Competitive rock climber (France, national level) · Overland travel across Africa · English, French (native), Spanish